General Derivation Strategy: Numerical Quadrature

To evaluate

$$\int_a^b f(x)\,dx$$

numerically, we replace the integrand $f(x)$ with an $(n+1)$ point Lagrange interpolating polynomial $P_n(x)$. The definite integral is approximated as:

$$\int_{x_0}^{x_n} f(x)\,dx = \int_{x_0}^{x_n} \sum_{k=0}^{n} L_k(x) f(x_k)\,dx + \int_{x_0}^{x_n} \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{k=0}^{n}(x - x_k)\,dx$$

The resulting expression

$$\sum_{i=0}^{n} a_i f(x_i)$$

is called a quadrature, where the weights a_i are determined by integrating the Lagrange basis functions:

$$a_i = \int_{x_0}^{x_n} L_i(x)\,dx.$$

1. Detailed Derivation: Trapezoidal Rule ($n = 1$)

Setup:

We use nodes $x_0 = a$ and $x_1 = a + h = b$, where the step size $h = b - a$.

Weight Derivations:

For a_0:

$$a_0 = \int_{x_0}^{x_1} \frac{x - x_1}{x_0 - x_1}\,dx = \frac{1}{-h}\left[\frac{x^2}{2} - x_1 x\right]_{x_0}^{x_1} = \frac{1}{-h}\left[\left(\frac{x_1^2}{2} - x_1^2\right) - \left(\frac{x_0^2}{2} - x_1 x_0\right)\right]$$

$$a_0 = \frac{1}{-h}\left[-\frac{x_1^2}{2} - \frac{x_0^2}{2} + x_1 x_0\right] = \frac{1}{-h}\left[-\frac{(x_1 - x_0)^2}{2}\right] = \frac{1}{-h}\left[-\frac{h^2}{2}\right] = \frac{h}{2}$$

For a_1:

$$a_1 = \int_{x_0}^{x_1} \frac{x - x_0}{x_1 - x_0}\,dx = \frac{1}{h}\left[\frac{x^2}{2} - x_0 x\right]_{x_0}^{x_1} = \frac{1}{h}\left[\left(\frac{x_1^2}{2} - x_0 x_1\right) - \left(\frac{x_0^2}{2} - x_0^2\right)\right]$$

$$a_1 = \frac{1}{h}\left[\frac{x_1^2}{2} - x_0 x_1 + \frac{x_0^2}{2}\right] = \frac{1}{h}\left[\frac{(x_1 - x_0)^2}{2}\right] = \frac{h}{2}$$

Error Term (Weighted Mean Value Theorem):

Since $g(x) = (x - x_0)(x - x_1)$ does not change sign on the interval $[x_0, x_1]$, we pull $f''(\xi)$ out of the integral:

$$E = \frac{1}{2} f''(\xi) \int_{x_0}^{x_1} \left(x^2 - (x_0 + x_1)x + x_0 x_1\right)\,dx$$

$$= \frac{1}{2} f''(\xi)\left[\frac{x^3}{3} - \frac{(x_0 + x_1)x^2}{2} + x_0 x_1 x\right]_{x_0}^{x_1} = -\frac{h^3}{12} f''(\xi)$$

Final Formula:

$$\int_{x_0}^{x_1} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_1)\right] - \frac{h^3}{12} f''(\xi)$$

Usage Explanation:

The Trapezoidal rule approximates the area as a single trapezium by connecting endpoints with a linear segment. It is exact for any function whose second derivative is zero, meaning polynomials of degree ≤ 1.

2. Detailed Derivation: Simpson's 1/3 Rule ($n = 2$)

Setup:

We use three nodes $x_0, x_1 = x_0 + h, x_2 = x_0 + 2h$ where the step size

$$h = \frac{b - a}{2}.$$

Weight Derivations:

For a_1 (Midpoint):

$$a_1 = \int_{x_0}^{x_2} \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}\,dx = \frac{1}{-h^2}\int_{x_0}^{x_2} (x^2 - (x_0+x_2)x + x_0x_2)\,dx$$

$$a_1 = -\frac{1}{h^2}\left[\frac{x^3}{3} - \frac{(x_0+x_2)x^2}{2} + x_0x_2x\right]_{x_0}^{x_2} = -\frac{1}{h^2}\left(-\frac{4}{3}h^3\right) = \frac{4h}{3}$$

For a_0 (Endpoint):

$$a_0 = \int_{x_0}^{x_2} \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}\,dx = \frac{1}{2h^2}\int_{x_0}^{x_2} (x^2 - (x_1+x_2)x + x_1x_2)\,dx$$

$$a_0 = \frac{1}{2h^2}\left[\frac{x^3}{3} - \frac{(x_1+x_2)x^2}{2} + x_1x_2x\right]_{x_0}^{x_2} = \frac{1}{2h^2}\left(\frac{2}{3}h^3\right) = \frac{h}{3}$$

By symmetry of equally spaced intervals,

$$a_2 = a_0 = \frac{h}{3}.$$

Final Formula:

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] - \frac{h^5}{90}f^{(4)}(\xi)$$

Usage Explanation:

Simpson's 1/3 rule fits a quadratic parabola through three points to capture curvature. It is far more accurate than Trapezoidal, achieving a degree of precision of 3, meaning it is exact for cubic polynomials.

3. Detailed Derivation: Simpson's 3/8 Rule ($n = 3$)

Setup:

We use four nodes x_0, x_1, x_2, x_3 where the step size

$$h = \frac{b-a}{3}.$$

Weight Derivations:

For a_0 (Outer weight):

$$a_0 = \int_{x_0}^{x_3} \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}\,dx = \frac{1}{-6h^3}\int_{x_0}^{x_3} (x-x_1)(x-x_2)(x-x_3)\,dx$$

Integrating the cubic expansion over the full interval $[x_0, x_0+3h]$ yields:

$$a_0 = -\frac{1}{6h^3}\left(-\frac{9}{4}h^4\right) = \frac{3h}{8}$$

For a_1 (Inner weight):

$$a_1 = \int_{x_0}^{x_3} \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}\,dx = \frac{1}{2h^3}\int_{x_0}^{x_3} (x-x_0)(x-x_2)(x-x_3)\,dx$$

Integrating this cubic term results in:

$$a_1 = \frac{1}{2h^3}\left(\frac{9}{4}h^4\right) = \frac{9h}{8}$$

By symmetry, the weights follow the pattern

$$a_3 = a_0 = \frac{3h}{8}, \qquad a_2 = a_1 = \frac{9h}{8}.$$

Final Formula:

$$\int_{x_0}^{x_3} f(x)\,dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right] - \frac{3h^5}{80}f^{(4)}(\xi)$$

Usage Explanation:

Simpson's 3/8 rule approximates the integral using a cubic polynomial fit over four nodes. While it shares the same degree of precision (degree 3) as the 1/3 rule, it is specifically required when data points are available in multiples of three sub-intervals.